

Yu Li

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Education

George Washington University	Washington, D.C.
<i>Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0</i>	<i>2025.08 – 2029.05</i>
Advisor: Prof. Tian Lan 🔗; Work with: Prof. Zhengling Qi 🔗	<i>(expected)</i>
Wuhan University, Hongyi Honor College	Wuhan, China
<i>B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0</i>	<i>2021.09 – 2025.06</i>

Experience

ByteDance Inc.	San Jose, CA
<i>Research Scientist Intern, TikTok Recommendation Core</i>	<i>2026.05 – 2026.08</i>
Owned the ranking stage of TikTok video recommendation, working on 3B Large Recommendation Model with MLA, time-aware RoPE, and mixed-query optimization.	

Research Highlights

My research develops principled reinforcement learning formulations for **LLM Reasoning**, spanning token-level credit assignment, preference optimization, and context conditioning [P1][P3][P4][P5], and extends these formulations to **Agentic RL** through hierarchical skill acquisition and multi-turn policy optimization [P2][ACL'26][INFOCOM'26]. On **Efficient AI**, my work focuses on parameter-efficient fine-tuning and long-context KV cache management [COLM'26][AAAI'26]. Earlier work on **Generative Models** covers diffusion-based personalized content-style synthesis [CVPR'26].

Preprints & Under Review

P=Preprint

- [P1] **OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning** 🔗
Yu Li, Rui Miao, Tian Lan, Zhengling Qi
- [P2] **ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning** 🔗
Yu Li, Rui Miao, Zhengling Qi, Tian Lan
- [P3] **Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO** 🔗
Yu Li, Tian Lan, Zhengling Qi
- [P4] **INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization** 🔗
Yu Li, Tian Lan, Zhengling Qi
 AI with Recursive Self-Improvement Workshop at ICLR, 2026
- [P5] **Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning** 🔗
Yu Li, Shu Hong, Tian Lan
Under review at EMNLP 2026, OA: 3.33/5

Selected Publications [\[Full List\]](#)

†=Equal Contribution

I have published **7** papers at top-tier conferences, with **5** of them as **first or co-first author**.

[COLM'26] MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference

Yu Li, Binxu Li, Tian Lan

[ACL'26 Findings] Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization

Yu Li, Sizhe Tang, Tian Lan

[CVPR'26] CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion

Yu Li, Yujun Cai, Chi Zhang

[ICASSP'26 Oral] KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models

Yu Li, Da Chang, Xi Xiao

[INFOCOM'26] ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense

Yu Li†, Sizhe Tang†, Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan

[ACM MM'26] Prompted Information Bottlenecks: Rethinking Layer-Wise Information Allocation in Frozen Vision Transformers

Yuqi Li, Xi Xiao, Yunbei Zhang, Lin Zhao, Yu Li, Aiden Zhao, Tianyang Wang, Hao Xu, Yingli Tian

[AAAI'26] Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT

Da Chang, Peng Xue, Yu Li, Yongxiang Liu, Pengxiang Xu, Shixun Zhang

Selected Awards & Honors

- **ICML 2026 Gold Reviewer**, top reviewers recognition. 2026
- **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. 2024
- **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. 2023, 2024
- **First-Class Scholarship**, Top 5%, ¥3,000, three times. 2022, 2023, 2024

Academic Services

- **Conference Reviewer:** 2027: AAAI. 2026: NeurIPS, ICML, ICLR, COLM, AAAI, ACM MM.
- **Journal Reviewer:** TPAMI, TMLR, ToN, Neurocomputing.